

END TERM EXAMINATION
THIRD SEMESTER [B.TECH] DECEMBER-2025

Paper Code: CIC-211**Subject: Object-Oriented Programming Using C++**

Time: 3 Hours
Maximum Marks: 60

Note: Attempt any five questions including Q. No. 1 which is compulsory. Select one question from each unit.

Q. No.	Question	Marks
1	Attempt any five questions: (a) Define the Object-Oriented Programming (OOP) paradigm. Discuss the evolution of C++ from C. (b) Explain the significance of member functions in encapsulation and abstraction. (c) What is an access specifier? Explain with the help of a suitable example. (d) What are the differences between data type and abstract data type? Give example. (e) Explain the concept of virtual functions and their role in achieving runtime polymorphism. (f) Define iterators and their significance in the STL.	4 × 5 = 20
UNIT – I		
2	(a) What is function prototyping? Why is it used in C++? Justify. (3 Marks) (b) Define inline functions in C++. (3 Marks) (c) (i) Discuss the concepts of call by reference and return by reference in C++. How do they differ from call by value? (ii) Write a program to find the greatest of two given numbers in two different classes using friend function. (4 Marks)	3 + 3 + 4 = 10
3	Discuss how C++ supports encapsulation, information hiding, inheritance, and polymorphism. Provide comprehensive examples to support your explanation.	10
UNIT – II		
4	(a) Explain Default constructor, Parameterized constructor, Dynamic constructor and Copy constructor with suitable code examples. (7 Marks) (b) What is operator overloading? Implement a class String containing member/friend functions to: (3 Marks) (i) Overload + operator to carry out the concatenation of strings. (ii) Overload = operator to carry out string copy. (iii) Overload <= operator to carry out the comparison of strings.	7 + 3 = 10

[P.T.O.]

Q. No.	Question	Marks
5	(a) What is the purpose of specifying a class in C++? How does a class declaration differ from a class definition? (4 Marks) (b) How does encapsulation contribute to the concept of abstract data types? Discuss the purpose and significance of a destructor. (3 Marks) (c) Explain the following terms: Garbage collection, Destructor methods. (3 Marks)	4 + 3 + 3 = 10
UNIT – III		
6	(a) Explain types of inheritance in detail. Explain Hybrid inheritance with a suitable program/example. (7 Marks) (b) Explain the advantages and limitations of each inheritance method. (3 Marks)	7 + 3 = 10
7	(a) How does function overriding contribute to polymorphism? (4 Marks) (b) What do you mean by polymorphism? Explain various types of polymorphism with suitable examples. (3 Marks) (c) Explain persistent objects and multiple inheritance in C++. (3 Marks)	4 + 3 + 3 = 10
UNIT – IV		
8	(a) What are containers in the STL? Describe the various components of the Standard Template Library in detail. (6 Marks) (b) Write a program in C++ to illustrate exception handling with multiple catch blocks. (4 Marks)	6 + 4 = 10
9	Write short notes on any two of the following: (5 + 5 = 10 Marks) (a) Containers in STL (b) Vectors (c) Exception Handling in OOP (d) C++ Stream Classes	5 + 5 = 10
